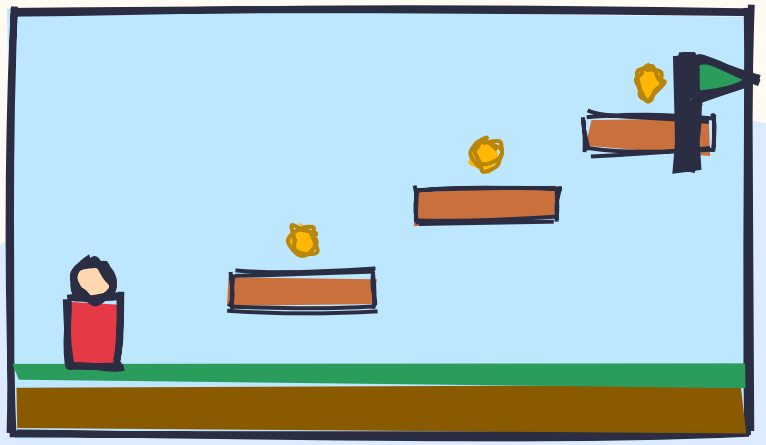
Junior Makers · Class 18

MakeCode Arcade: My First Platform Game

Build a real game you can play on any computer

Sprites | Gravity & jumping | Events | Level design



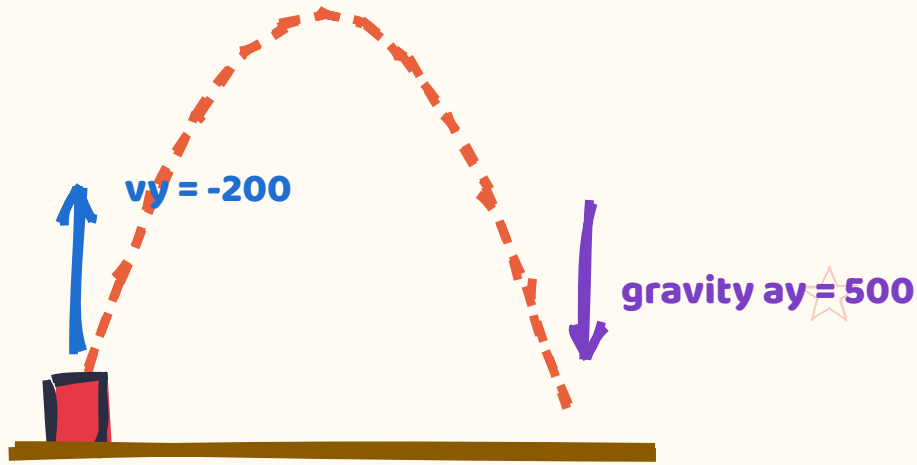
Did you know?
Super Mario Bros (1985) is one of the best-selling games ever. Its first level was designed to teach you the rules without a single word.

Let's think...

Why does Mario fall down, not up?

Game physics

When a game character jumps, it goes up, slows down, and falls back. Nobody draws each step by hand. The game uses **rules**: a bit of maths called **physics**. Today you write those rules yourself!



It's a video?

An animator draws it?

Code + physics!

Today: you build a **platform game** in MakeCode Arcade with jumping, coins, a score and a finish flag.

Learning objectives

By the end of this class I can...

1	**Sprites:** make a **player sprite** and move it with the controller	**Proof:** my player walks left and right
2	**Gravity & jump:** use **gravity** (ay) and **velocity** (vy) to make a jump	**Proof:** my player falls, lands and jumps
3	**Events:** use **events**: button pressed and sprites overlapping	**Proof:** collecting a coin adds 1 point
4	**Level design:** design a level, **playtest** it and make it fairer or more fun	**Proof:** one change I made after a friend played

LEARN

Objective 1 · Sprites · video

Video time!



Invention of the Video Game

The Dr. Binocs Show (Peekaboo Kidz) · about 7 minutes · youtu.be/VlhlGk_9X4A

Watch for...

1. What was one of the **first** video games?
2. Why did early games use simple shapes and pixels?
3. How have games changed since then?

Scan to watch

After the video: Tell your partner your favourite game. Is it a platform game? What makes it fun?

Objective 1 · Sprites

Sprites live on an x, y grid

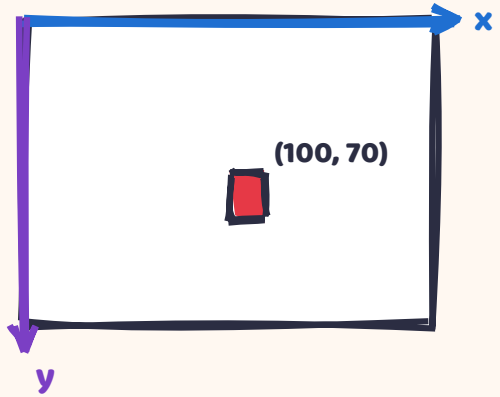
The idea

A **sprite** is any picture that moves: the player, coins, enemies.
The screen is **160 × 120 pixels**.

- **x** goes left → right
- **y** goes **top** → **bottom** (so bigger y = lower!)

Velocity (vx, vy) is how fast a sprite moves each second.

Example: a chess board



On a chess board every square has a name, like E4. Tell a friend 'knight to E4' and they know exactly where. x and y are the same idea for a game screen.

Objective 2 · Gravity & jump

Gravity pulls, jumps push

The idea

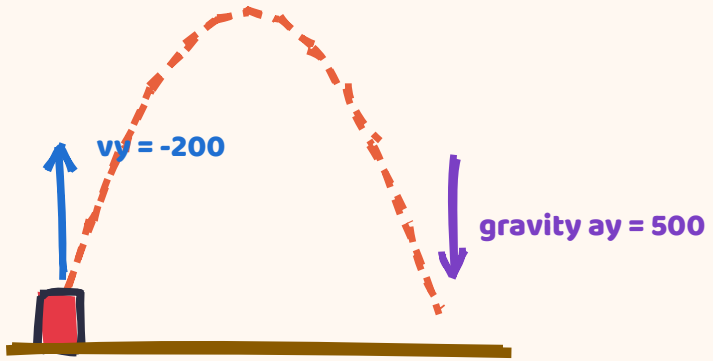
ay = 500 means: every second, the downward speed grows by 500. That is **gravity**.

To jump, give the player a big **upward** speed: **vy = −200** (negative = up). Gravity slows it, stops it, and pulls it back down. That makes a smooth curve!

Moon mode

Change 500 to 200: you're on the Moon! Change it to 1000: heavy planet. Experiment!

Example: throwing a ball



Throw a ball straight up. It's fastest when it leaves your hand, slows down, stops for a moment at the top, then falls faster and faster. Gravity changes its speed all the time.

Objective 3 · Events

Events: when this happens, do that

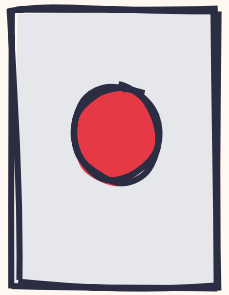
The idea

Games are full of **events**:

- **on A button pressed** → jump
- **on sprite overlaps** a coin → add 1 point, remove the coin
- **on overlap tile** flag → you win!
- **on overlap tile** lava → lose a life

The game keeps checking for events many times a second.

Example: a doorbell



DING!

A doorbell does nothing all day. The moment someone presses it, it rings. Event blocks wait in the same way, and act the moment their event happens.

Real world

Who makes games?



Artists
Draw sprites and backgrounds



Level designers
Plan where platforms, coins and dangers go



Programmers
Write the rules: physics, events, scores



Testers
Play again and again to find bugs and boring bits

Pakistan has a growing game industry. Which of these four jobs would you like?

Made in Pakistan
Studios in Lahore and Karachi make mobile games played by millions around the world.

Objective 4 · Level design

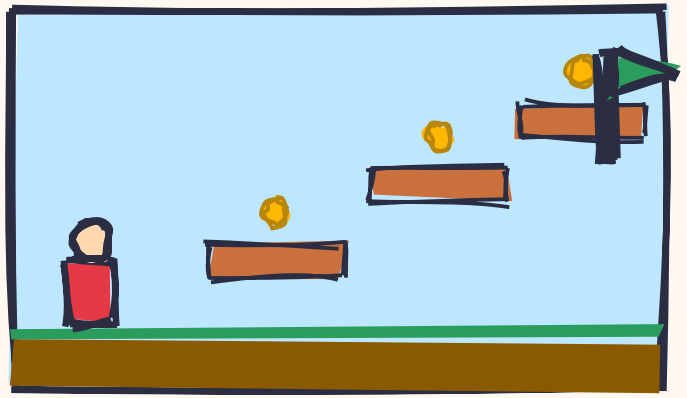
Good levels teach the player

The idea

Great level designers follow simple rules:

- **Start easy**: the first jump is small and safe.
- **Show, then test**: show a coin on an easy jump before a hard one.
- **Guide the eye**: coins make a trail to follow.
- **Fair danger**: hazards are visible, never hidden.

Example: learning to ride a bike



Nobody starts on a mountain path. You start on flat ground, with someone holding the seat, then try small slopes. A good level does the same: easy first, harder later.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human platformer
2. Player, level and gravity
3. Make it jump
4. Coins and score
5. Win, lose and lives
6. Swap and play
7. Bug hunt

Class game

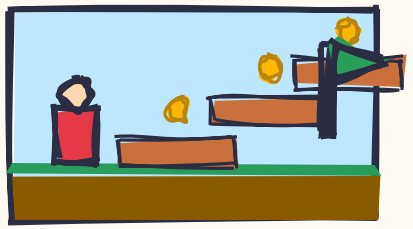
Human platformer

How to play

1. Make a 'level' in the classroom with tape: floor, two 'platforms' (chairs are NOT allowed to stand on!), and 3 paper coins.
2. The **player** may only do what the **game** (a partner) says: **LEFT**, **RIGHT**, **JUMP**.
3. Every JUMP goes up 2 tiles, then gravity brings you down.
4. Collect all coins and reach the flag. Count the moves.

Then discuss

What rules did the 'game' follow? Write them as: **when ___ happens, do ___.**



Big idea

A game is a set of rules that react to what the player does.

Objective 1 · Sprites · Build step 1 of 4

Player, level and gravity

```
on start
    set tilemap to level1
    set mySprite to sprite 🧍 of kind Player
    controller move mySprite with buttons vx 100 vy 0
    camera follow sprite mySprite
    set mySprite ay (acceleration y) to 500
```

vy is 0!
We only let the controller move left-right. Up and down is gravity's job.

Open it

Go to **arcade.makecode.com** → New Project. Blocks are on the left; the game simulator is on the left edge.

Tilemap

Click the grey square in 'set tilemap' to draw your level: **walls** are solid, other tiles are background.

Objective 2 · Gravity & jump · Build step 2 of 4

Make it jump

```
on A button pressed
    if mySprite is hitting wall bottom then
        set mySprite vy (velocity y) to -200
```

Why the if?

Without it, pressing A again in mid-air lets you fly forever! Only jump when your feet are **touching the ground**.

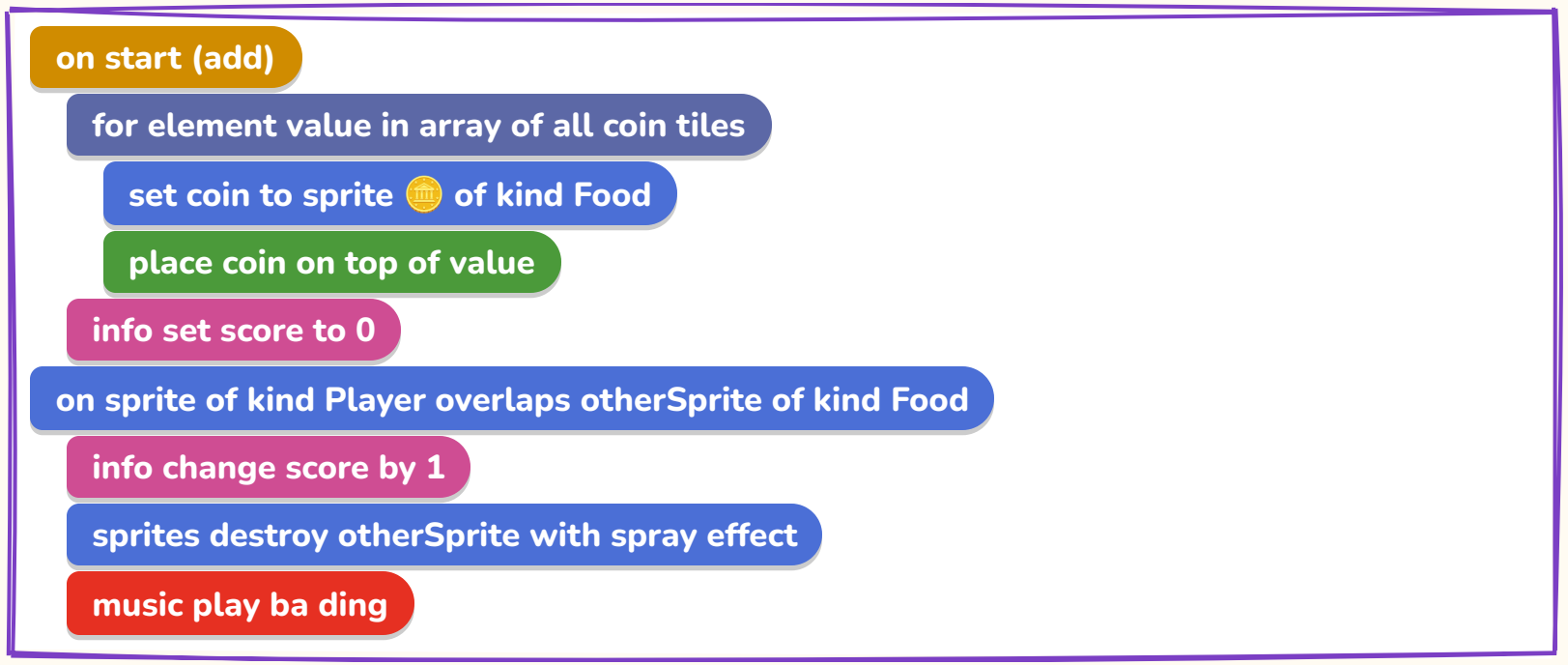
Settings	Jump height	Feels…
ay 500, vy −200		
ay 500, vy −300		
ay 300, vy −200		

Test it

Find settings where you can JUST reach the highest platform. That's good level design!

Objective 3 · Events · Build step 3 of 4

Coins and score



Kinds

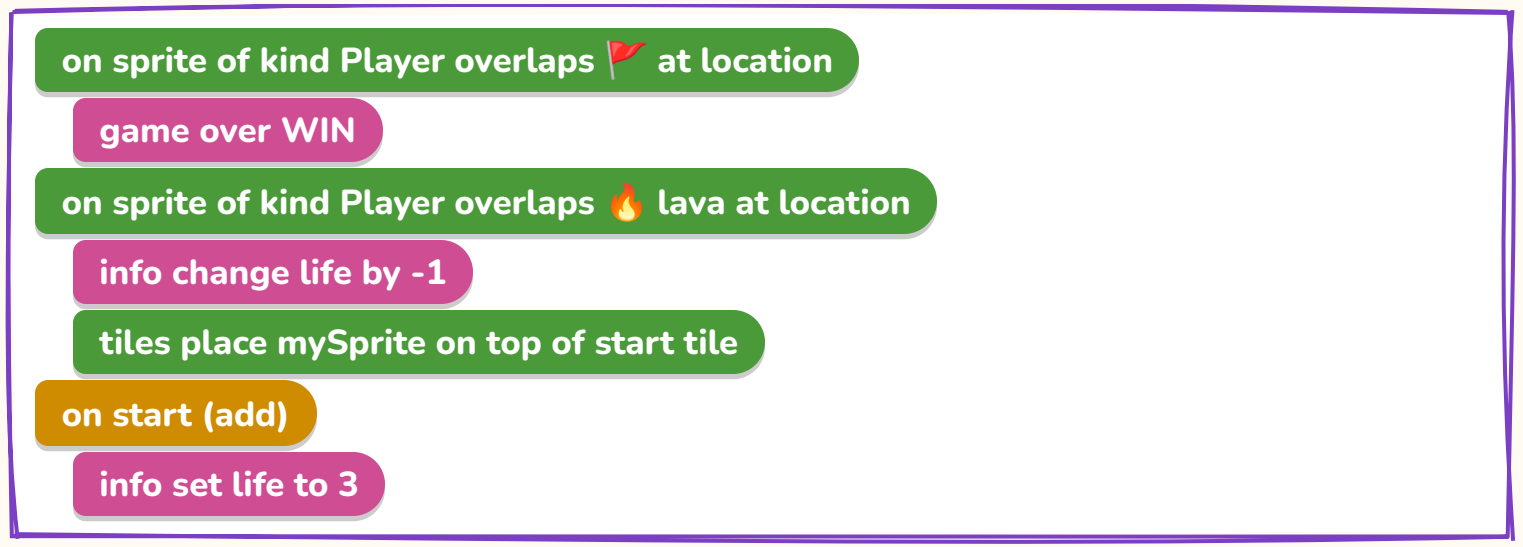
Player, **Food**, **Enemy** are 'kinds'. The overlap event checks which kinds touched.

Place coins

Draw a special coin tile in your tilemap, then the loop turns each one into a coin sprite.

Objective 3 · Events · Build step 4 of 4

Win, lose and lives



Fair danger

Put lava where a careful player can **see** it coming. Surprise traps feel unfair.

When lives hit 0

MakeCode ends the game automatically and shows 'Game Over'.

Objective 4 · Level design · playtest

Swap and play

Tester card	Answer
Could you finish? How many tries?	
Hardest part	
Most fun part	
Anything unfair?	
One idea for the maker	

Rules for testers

Don't tell the maker what to do while playing. Just play and think out loud: 'oh, I didn't see that!'

Makers

Watch silently. Write down where your tester got stuck. That's your to-do list!

Test like an engineer

Bug hunt

Bug	Likely cause	Did it happen?
Player falls through the floor	floor tiles not set as walls	
Player flies away when A is pressed	missing 'hitting wall bottom' check	
Coins don't disappear	destroy otherSprite missing	
Camera doesn't move	camera follow sprite missing	
Score goes up by 2	two overlap blocks for the same thing	

Debug checklist

Use the slow-motion button (the snail) in the simulator to watch exactly what happens.

Change ONE thing, then play again. That way you know what fixed it.

Level up!

Game designer challenges

Choose your challenge



Enemy: an enemy that walks back and forth on a platform (vx and 'bounce on wall').



Level 2: when you reach the flag, load a second tilemap.



Timer: info start countdown 60. Can you finish in time?



Sound design: different sounds for jump, coin and lava.

Think like an engineer

Why do most platform games make the first level easy?

Boss level

Download your game to a handheld or share the link with your family!

Share it

Your game, on any device

Share link

Click **Share** → **Publish** to get a link. Anyone can play it in a browser, even on a phone.

Safe sharing

Use a game name, not your full name. Only share the link with family, friends and teachers.

Arcade night

Collect all the links on one page and hold a class arcade: everyone plays everyone's game!

Objective check

Can I really do it? Prove it.

01	Show your player moving left and right. Which block does that?	○ ○ ○
02	What do ay = 500 and vy = −200 do? Show your jump.	○ ○ ○
03	Show a coin being collected. Which event block made the score go up?	○ ○ ○
04	Tell us one change you made after a friend tested your level.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Class arcade: play 3 other games and give each a tester card.

Show what you know

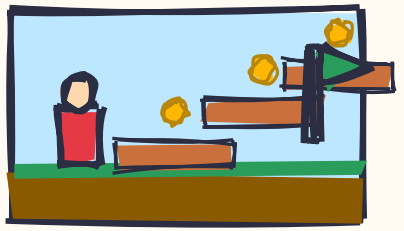
What did we learn?

- [x] Sprites move on an x, y grid (y goes DOWN)
- [x] Gravity (ay) changes speed; a jump is a big upward vy
- [x] Events: when this happens, do that
- [x] Kinds let the game tell coins from enemies
- [x] Good levels start easy and are fair

Exit ticket: why do we check 'hitting wall bottom' before jumping?

At home

Play a platform game at home and count how many rules (events) you can spot.



Next class:
Binary Bracelets & Secret Codes!